Mahmoud Alyosify

AI Agent Systems Engineer | Multi-Agent Orchestration, Tool Use & Workflow Automation
MSc Artificial Intelligence, Queen's University (Canada)

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PROFESSIONAL SUMMARY

AI Agent Systems Engineer building **autonomous, multi-step systems that do real work** — not chat wrappers. Currently an **AI Engineer Intern at Injaz Tech**, building a **LangGraph** corrective-RAG agent with routing, evidence grading, query rewriting and hallucination self-checks behind a FastAPI service. Architected **HORUS Sentinel**, a swarm of specialised agents that continuously collect open-source intelligence and correlate every finding into a **living knowledge graph**, over which a self-hosted fine-tuned LLM produces **evidence-backed reports**, compressing hours of analyst work into minutes. Fluent in **LLM API integration** (prompt design, **structured outputs**, **tool / function calling**), **RAG**, and end-to-end agentic automation pipelines. Also studies agents empirically — a GitHub-mining project quantifying **review effort and failure modes of agent-generated pull requests**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, researching token efficiency — the cost dimension that decides whether agent loops are economically viable.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Agentic Systems	Multi-Agent Architectures, Agent Orchestration, Think–Plan–Act Loops, Tool / Function Calling, Structured Outputs, Autonomous Task Decomposition, Workflow Automation, LangChain, LangGraph, Corrective RAG, Bounded Self-Correction Loops, Knowledge-Graph Grounding
LLM Integration	LLM APIs (Anthropic, OpenAI), Ollama, Prompt Engineering, RAG, Vector Search (FAISS, ChromaDB), Cross-Encoder Re-Ranking, Fine-Tuning (LoRA, QLoRA, PEFT), Self-Hosted Inference, Hugging Face Transformers
Agentic Coding Tools	Claude Code, Cursor, GitHub Copilot
Engineering	Python, TypeScript, FastAPI, Pydantic, pytest, Docker, AWS, ONNX, CI/CD (GitHub Actions), MongoDB, MySQL, Git/GitHub, Linux
Evaluation	LLM-as-Judge, BERTScore, ROUGE, Token-Cost & Latency Profiling, Statistical Testing, Reproducible Experiment Design
Programming	Python, TypeScript, JavaScript, SQL, C++, C#, C, Go, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built a **LangGraph** corrective-RAG agent: embedding-similarity plus LLM routing, retrieval with cross-encoder re-ranking, relevance grading backed by quoted evidence, query rewriting and a hallucination self-check, with every loop bounded (worst-case termination proven by tests).
- Implemented routing and grading decisions as typed **Pydantic** structured outputs verified in code; the grader redesign lifted its recall from **17% to 88%** and cut false refusals from **5/20 to 1/20**.
- Exposed the agent as a **FastAPI** service with SSE streaming of node status and tokens, per-session memory, caching, retries, a circuit breaker, a hard per-request LLM-call budget and prompt-injection guardrails (**0** prompt leaks across 8 injection probes).

Machine Learning Instructor — *Aug 2026 – Sep 2026*
National Telecommunication Institute (NTI) — Arish & Assiut, Egypt

- Rehired to teach two **90-hour** applied ML bootcamps (**50 trainees**) at the Arish (online) and Assiut branches — regression, SVM, KNN, Naive Bayes, PCA and neural networks with Python labs, final exams and capstone-project evaluation.

Machine Learning Instructor — *Jul 2025 – Sep 2025*
National Telecommunication Institute (NTI) — Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED AGENTIC AI PROJECTS

HORUS Sentinel — Autonomous Multi-Agent OSINT & Threat-Intelligence Platform *2026*
Python · Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected an autonomous intelligence analyst: a **swarm of specialised passive agents**, each owning one collection domain, runs continuously and independently rather than under a single monolithic prompt.
- Designed a **living Intelligence Knowledge Graph** as the shared memory layer, so every agent's finding is correlated with prior evidence instead of being lost between runs.
- A **self-hosted, fine-tuned language model** reasons over that graph to deliver **prioritised, evidence-backed reports** — every conclusion traceable to its source — compressing hours of manual analyst work into minutes.

Automated Multimodal Agent — PDF to Narrated Presentation *2026*
Agentic Orchestration · Multimodal Generation · Python

- Built an autonomous **think–plan–act** pipeline that converts a source PDF into a fully narrated PowerPoint deck, coordinating document understanding, structured slide generation and audio narration as distinct tool-using stages.
- Removed an entirely manual workflow end to end — the agent decides structure and content, not just formatting.

Analyzing Review Effort: Human vs. Agentic Pull Requests *2026*
Empirical Study · GitHub Mining · Agent Evaluation

- Built a reproducible GitHub-mining pipeline quantifying **review effort and recurring failure modes** of AI-agent-generated pull requests versus human-authored ones across large open-source repositories — evidence about where coding agents actually break down.

Verbosity-Aware Decoding — Token Economics for Agent Loops *2026 – Present*
MSc Research Project (CISC 898), Queen's University · PyTorch · AWS GPU

- Designed a **training-free LogitsProcessor** that cuts generated tokens at decoding time under an **entropy-driven gate** — directly addressing the cost multiplier that makes long multi-agent loops expensive — validated across three model families (**25–35%** fewer tokens on MT-Bench) with latency and GPU-energy measurement.

Additional Projects

- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records and deployed it on **AWS** as an interactive assistant.
- **SimCLR-Vision-SSL** — contrastive representation learning with ONNX/FAISS retrieval — the vector-search layer behind RAG-style agent memory.
- **Also:** SCRAP · AudioShield · RL-Cloud-Autoscaler (PPO/DQN) · Vitalism Solution (rPPG).

CERTIFICATIONS & AWARDS

Generative AI & Prompt Engineering — IBM / Coursera

Machine Learning Specialization — DeepLearning.AI

Digilians Presidential Scholarship — Queen's University

Semi-Finalist — Microsoft Imagine Cup 2023

AWS Academy — ML Foundations, ML for NLP, Data Engineering

HCIP – AI (Huawei)

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.